https://github.com/coolhero/spec-kit-skills

```
auth
      smart-sdd/concerns/auth.md
      smart-sdd/archetypes/ai-assistant.md
      reverse-spec/concerns/auth.md
      foundations/react.md

      specify
      implement
      verify

      →
      →
      →
      →
```

```
--learn
```

localStorageUser

spec-kit-skills

npm run build

{mcpMode, mcpServers}{mcpEnabled, servers}

tool(){execute: async () => {...}}{type: "mcp", serverId: "xxx"}

```
UNDERSTAND:  /code-explore   orient -> trace -> synthesis
ANALYZE:     /reverse-spec   scan -> classify -> extract -> generate
BUILD:       /smart-sdd      init -> add -> specify -> plan -> tasks
                              -> analyze -> implement -> verify -> merge
EXTEND:      /domain-extend  browse -> detect -> extend -> validate
```

/domain-extend

→→→→

add

```
        P1. Context Continuity (what to protect)
       /          |           \
Domain Profile  Source Code   Cross-Feature
full pipeline   preservation  GEL memory
      \           |           /
     P2. Enforce, Don't Reference (how to protect)
                  |
     P3. File over Memory (where to store)
```

addspecifyverify

gui + realtimespecifycli + resilience

```
init
specify
add
implementverify
```

ai-assistantgui

_resolver.mdguirealtime

```
gui + realtimespecifyplanimplement
```

ModelSelectorimplement

pre-context.mdspec.md

pre-context.md

entity-registry.md

api-registry.md

sdd-state.md

roadmap.md

pre-context.md

id, email, displayName, roleuserId, name, userTypeusername, permissions

MANDATORY

```
WRONG: Run build+TS, display "verify complete", proceed to merge
   -> 10 SCs unverified. Feature unreachable from UI.

RIGHT: Read verify-phases.md -> execute Phase 0-4 ->
```

```
    show SC coverage matrix -> AskUserQuestion
```

```
pipeline.md:1453injection/implement.md:598
```

console.log(agent.memory)

git diff

sdd-state.md

```
git log
```

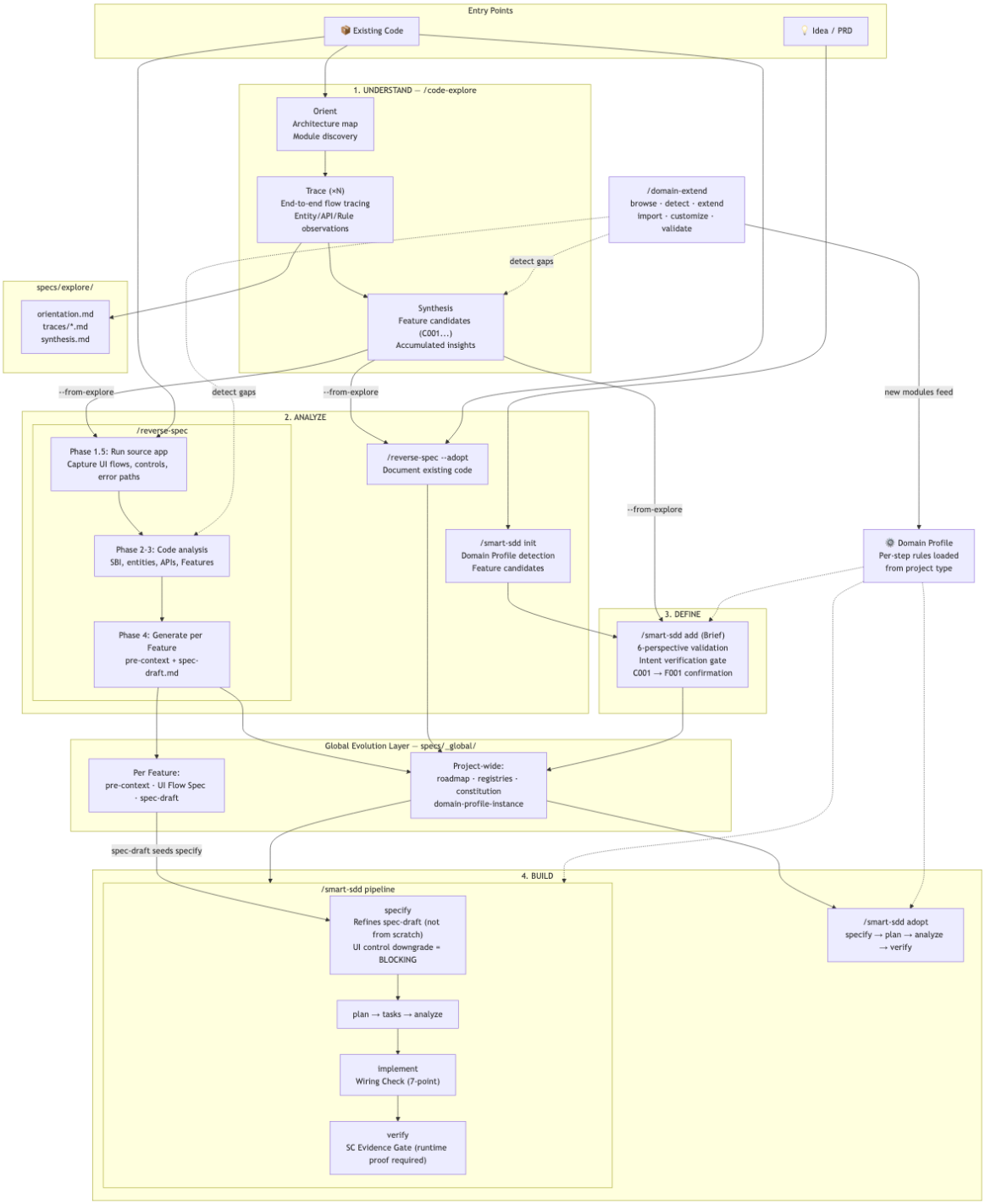
/code-explore/smart-sdd

```
WRONG: "Playwright limitation -> drag-and-drop skip"
  -> Untested. Bug ships silently.

RIGHT: AskUserQuestion "Please drag a file onto the upload zone.
       Does the status change from 'idle' to 'processing'?"
  -> User verifies. Result documented.

WRONG: "API key absent -> external-dep -> skip"
  -> Feature untested in real conditions.

RIGHT: AskUserQuestion "Please set your API key in Settings > Provider.
       After setup, I will auto-test embedding generation."
  -> User provides key. Agent tests. Real verification.
```


Entry Points
📦 Existing Code
💡 Idea / PRD
1. UNDERSTAND — /code-explore
Orient
Architecture map
Module discovery
Trace (×N)
End-to-end flow tracing
Entity/API/Rule
observations
/domain-extend
browse · detect · extend
import · customize ·
validate
detect gaps
specs/explore/
orientation.md
traces/*.md
synthesis.md
Synthesis
Feature candidates
(C001...)
Accumulated insights
--from-explore
detect gaps
--from-explore
new modules feed
2. ANALYZE
/reverse-spec
Phase 1.5: Run source app
Capture UI flows, controls,
error paths
/reverse-spec --adopt
Document existing code
--from-explore
Phase 2-3: Code analysis
SBI, entities, APIs, Features
/smart-sdd init
Domain Profile detection
Feature candidates
🧠 Domain Profile
Per-step rules loaded
from project type
3. DEFINE
Phase 4: Generate per
Feature
pre-context + spec-
draft.md
/smart-sdd add (Brief)
6-perspective validation
Intent verification gate
C001 → F001 confirmation
Global Evolution Layer — specs/_global/
Per Feature:
pre-context · UI Flow Spec
· spec-draft
Project-wide:
roadmap · registries ·
constitution
domain-profile-instance
spec-draft seeds specify
4. BUILD
/smart-sdd pipeline
specify
Refines spec-draft (not
from scratch)
UI control downgrade =
BLOCKING
plan → tasks → analyze
implement
Wiring Check (7-point)
verify
SC Evidence Gate (runtime
proof required)
/smart-sdd adopt
specify → plan → analyze
→ verify

→

```
              Domain Profile
              |
   INTERFACE --| What does the app expose?    -> gui, http-api, cli, ...
   CONCERN ----| What cross-cutting patterns? -> auth, async-state, ipc, ...
   ARCHETYPE --| What domain philosophy?      -> ai-assistant, microservice, ...
   FOUNDATION -| What framework constraints?  -> React, Electron, Go, ...
   CONTEXT ----| What project situation?       -> Mode + Scale + Modifiers
```

guihttp-apiclituidata-iomobilelibraryembeddedgrpck8s-api

/smart-sdd init/reverse-spec

---

realtimeai-assistant

```
# Universal principle
"Isolate side effects from render logic"

# Framework examples
React:   useEffect for side effects
Vue:     watch/watchEffect
Django:  signals or middleware
Go:      goroutine with context.Context
```

useEffect

greenfieldrebuildincrementaladoption

prototypemvpproduction×solosmall-teamlarge-team

+migration+compliance

prototype + soloproduction + large-team

_resolver.md

```
Profile:     desktop-app
Interfaces:  [gui]                    <- Axis 1
Concerns:    [async-state, ipc, external-sdk, realtime]  <- Axis 2
Archetype:   ai-assistant             <- Axis 3
Foundation:  electron                 <- Axis 4
Context:     rebuild / mvp / solo     <- Axis 5 (mode / scale)
```

_core.md

interfaces/gui.md

concerns/async-state.md

concerns/ipc.md

concerns/external-sdk.md

concerns/realtime.md

archetypes/ai-assistant.md

foundations/electron.md

contexts/modes/rebuild.md

→

→

→

→

→

×

domain-profile-instance.md

specs/_global/domain-profile-instance.md

/add/add/specify

```
## 2. Per-Concern Decisions

### auth
| Question (S5)        | Answer                        | Feature | Date       |
|----------------------|-------------------------------|---------|------------|
| Auth method?         | Server-side sessions + Redis | F003    | 2026-03-25 |
| Token expiry?        | 1h, sliding window renewal   | F003    | 2026-03-25 |
| Authorization model? | RBAC (admin/member/viewer)   | F003    | 2026-03-25 |
| Concurrent sessions? | 1 per device, new login      | F003    | 2026-03-25 |
|                      | expires previous              |         |            |

### token-budget
| Question (S5)        | Answer                        | Feature | Date       |
|----------------------|-------------------------------|---------|------------|
| Budget unit?         | Org monthly + User daily     | F004    | 2026-03-25 |
| Over-limit behavior? | 429 response + dashboard     | F004    | 2026-03-25 |
|                      | alert                         |         |            |
| Reset policy?        | Monthly auto-reset, no       | F004    | 2026-03-25 |
|                      | carry-over                    |         |            |
```

××

```
init
specify
parity
verify
_core.md
add
implementgui
planimplementverify
verify
add
```

```
/smart-sdd add "user authentication"
```

```
auth.md
ai-assistant
auth + realtime
```

domain-profile-instance.mdspecs/F00N/pre-context.md

/adddomain-profile-instance.md

→

```
react.mdspring-boot.md_foundation-core.md
```

auth

smart-sdd/concerns/auth.md

```
authauthenticationloginJWTOAuthsessionpassport
/init
/specify
/add
/plan/implement/verify
```

smart-sdd/archetypes/ai-assistant.md

```
/init
```

```
/specify
```

reverse-spec/concerns/auth.md

```
passportnext-auth@auth0/*express-session
**/middleware/auth.***/guards/***/strategies/*
```

→→→→

auth

→→→→→

```
B001: POST /auth/login with valid credentials -> 200 + session cookie  [P1]
B002: POST /auth/login with wrong password -> 401 + error message      [P1]
B003: Access protected route without token -> 302 redirect to login    [P1]
B004: Access admin route as regular user -> 403 forbidden              [P1]
B005: Session expires after 1 hour -> next request redirects to login  [P2]
B006: Logout -> session invalidated, cookie cleared                    [P1]
B007: Concurrent login from second device -> first session preserved   [P3]
```

foundations/react.md

---

useMemoProtectedRoute

foundations/django.mdpytestfactory_boy

specify

/specify

×

SC Merge at specify

```
Final SC list for F-002 (Auth):
  SC-001: Login success         <- S1 (auth concern)
  SC-002: Login failure         <- S1 (auth concern)
  SC-003: Token expiry + refresh <- S1 (auth concern)
  SC-004: Protected route 401   <- S1 (auth concern)
  SC-005: Protected route 403   <- S1 (auth concern)
  SC-006: LLM call with session <- A2 (ai-assistant archetype)
  SC-007: Re-auth on reconnect  <- cross-concern (auth x realtime)
```

implement

/implement

useMemoProtectedRoute

verify

npm run testnpm run build

rebuild+migration

specify
implement
verify

```
+migration

+migration

+migration
  →


http-api + auth + data-integrity | microservice archetype
| java-spring foundation | rebuild + production×small-team
| +migration



CONNECT BYDECODENUMBER(38)

+migration

+migrationDECODE→CASE WHENNULLCONNECT BY

spring.datasource→NUMBER→DECIMALNUMBER(38)NUMERICNVL→COALESCENVLCOALESCE+migration


+migration
  →


http-api + grpc + auth + resilience + message-queue
| microservice archetype | spring-boot foundation
| rebuild + production×large-team | +migration




+migration

→+migration

+migration

+migration
  →


http-api + graphql + auth + async-state
| public-api archetype | typescript + apollo-server foundation
| incremental + mvp×small-team | +migration




+migration
```

+migration

+migration

+migration

```
→→→→+migration+migration
```

→

http-api + auth + async-state | microservice | spring-boot | rebuild + production×large-tea
| +migration

pom.xmlbuild.gradlesourceCompatibility = '1.8'Optional+migration

DateCalendarjava.timesun.misc.Unsafe--add-opensjavax.*jakarta.*

→→

+migration

+migration→→

String.trim()String.strip()Pattern.matches()HashMap

jakarta.*@ConstructorBindingspring.config.importvarpermits

+migrationnew Date()→Instant.now()

mvn clean verify -Dmaven.compiler.release=21

+migration

```
HashMap+migration
```

specifyimplement

_core.md

interfaces/{name}.md

concerns/{name}.md

archetypes/{name}.md

foundations/{framework}.md

org-convention.md

specs/domains/

contexts/modes/{mode}.md

domain-custom.md

```
specs/domains/token-budgetai-gatewayorg-convention.md
git pull
```

gui + auth + realtime + ai-assistant + electron + rebuild

```
Step 1: _core.md S1          -> 12 universal SC rules loaded
Step 2: gui.md S1            -> "Every form needs validation SC" + 8 rules
Step 2: http-api.md S1       -> (not loaded -- gui project, not http-api)
Step 3: auth.md S1           -> "Login/logout/token SC required" + 6 rules
Step 3: realtime.md S1       -> "Streaming lifecycle SC required" + 5 rules
Step 4: ai-assistant.md A2   -> "Provider abstraction SC" + 4 extensions
Step 4b: electron.md F2      -> "IPC bridge, main/renderer SC" + 3 constraints
Step 5: (no org-convention)  -> skipped
Step 6: rebuild.md           -> "Source parity SC required"
Step 6b: specs/domains/      -> (no project-local modules)
Step 7: (no domain-custom)   -> skipped

Result: 38 rules from 6 modules + 1 context mode
```

```
         (vs. 200+ rules if ALL modules loaded = 81% context savings)
```

cli + resilience + microservice + go

~

~

~×~

~

~

~

gui.mdrealtime.mdai-assistant

guirealtime

```
_core.md_foundation-core.mdverify-build-test.md


     eslint.config.js

_foundation-core.mdfoundations/nestjs.md

     ruffeslint.config.js
     .prettierrc
```

```
_foundation-core.mdnpx tsc --noEmitmypygo vetcargo check
```

```
Level 0: Build + TS + Lint          (syntax/type errors)
Level 1: + App launches without crash (import/config errors)
Level 2: + UI elements render        (CSS/template errors)
Level 3: + Interactions correct state (logic errors)
Level 4: + Data persists across restart (round-trip errors)
```

→→→→→

```
grep -c "Guard [1-7]" injection/*.md verify-phases.md pipeline.md
```

domain-profile-instance.md
/add/specify

constitution-seed.md
/init/reverse-spec.specify/memory/constitution.md

coverage-baseline.md

verify-report.md
specs/NNN-feature/verify-report.md

demos/F00N-name.sh
--ci

demos/DG{N}-scenario.sh

entity-registry.mdapi-registry.mdsdd-state.mddomain-profile-instance.md

```
F001 (Auth):
  specify -> entity-registry += User{id, email, displayName, role}
  specify -> api-registry += POST /api/auth/login {...}
  implement -> code written

F002 (Task CRUD):
  specify -> reads entity-registry -> sees User entity
```

```
  specify -> SC: "task.owner references User.id" (not a new User)
  specify -> api-registry += GET /api/tasks {...}
  implement -> imports User type from F001

F003 (Dashboard):
  specify -> reads both registries
  specify -> SC: "dashboard displays User.displayName"
  specify -> SC: "dashboard lists tasks via GET /api/tasks"
  plan -> Integration Contract: consumes User from F001, Tasks from F002
  implement -> references both existing APIs
  verify -> Phase 4 (cross-Feature): dashboard + auth + tasks together
```

User.orgId

```
## User
- Owner: F003-auth-multi-tenancy
- Fields:
  - id: string (UUID)
  - email: string (unique)
  - role: enum (admin | member | viewer)
  - orgId: string (FK -> Organization)
  - passwordHash: string
- Used by: F003, F004-token-budget, F005-prompt-guard
- Notes: F004 reads User.orgId for budget lookup

## Organization
- Owner: F003-auth-multi-tenancy
- Fields:
  - id: string (UUID)
  - name: string
  - plan: enum (free | pro | enterprise)
  - monthlyTokenBudget: number
  - usedTokens: number
- Used by: F003, F004
```

```
## POST /auth/login
- Provider: F003-auth-multi-tenancy
- Request: { email: string, password: string }
- Response 200: { sessionId: string, user: User }
- Response 401: { error: "INVALID_CREDENTIALS" }
- Auth: None (public endpoint)
- Consumers: F002-llm-gateway (post-login redirect)

## POST /v1/chat/completions
- Provider: F002-llm-gateway
- Request: { model: string, messages: [], stream: boolean }
- Response 200 (stream): SSE events { delta, done }
- Response 429: { error: "TOKEN_LIMIT_EXCEEDED" }
- Auth: Bearer session token
- Consumers: Frontend
```

```
- Notes: F004 added 429 response; F005 adds input filtering
```

/specifyUser.orgId

sdd-state.md

```
# SDD State

**Project**: My Knowledge Base App
**Origin**: rebuild
**Domain Profile**: desktop-app
**Interfaces**: gui
**Concerns**: async-state, ipc, external-sdk, realtime
**Archetype**: ai-assistant
**Framework**: electron
**Context Mode**: rebuild
**Context Scale**: mvp x solo
**Context Modifiers**: none
**Artifact Language**: en
**Source Path**: /Users/dev/original-app
**Scope**: core
**Active Tiers**: T1
**Constitution Version**: 1.0
**State Schema Version**: 2.0
```

```
| FID  | Name            | Step      | Status    | Branch            |
|------|-----------------|-----------|-----------|-------------------|
| F001 | Auth            | verify    | active    | 001-auth          |
| F002 | Knowledge Base  | specify   | queued    | ---               |
| F003 | Chat            | ---       | planned   | ---               |
```

reference/injection/{command}.md

Type "continue" to review the results.

injection/constitution.md
injection/specify.md
injection/plan.md
injection/tasks.md
injection/analyze.md
injection/implement.md
injection/verify.md
injection/parity.md
injection/adopt-specify.md
injection/adopt-plan.md
injection/adopt-verify.md

specify

_core.md

gui.md

realtime.md

ai-assistant

→

→

speckit-specify

code-explorereverse-specsmart-sdddomain-extend

/code-explore

/code-explore /path/to/project

package.jsongo.modCargo.toml

@KafkaListener.proto

specs/explore/orientation.md

→→→→

→→→→→→→

```
→→

go func()pool.submit()ActorRef
```

/smart-sdd add --from-explore

specs/

add --to

--learn

--learn

specs/explore/learn/synthesis.md

--no-branch

```
--learn
```

User{id, email}User{id, role}User{id, email, role}

/reverse-spec

analyze.md

analyze-scan.md

analyze-runtime.md

analyze-deep.md

analyze-classify.md

roadmap.mdentity-registry.mdapi-registry.mdconstitution-seed.mdanalyze-generate.md

```
B001: User can log in with email and password
B002: Failed login shows inline error message after 1 second
B003: 3 failed attempts triggers 15-minute lockout
B004: Session token stored in httpOnly cookie, expires in 24h
```

spec-draft.mdspecify

reverse-spec --from-explore

```
/code-explore/reverse-spec --from-explore

--adopt
     --scope full
     --stack same

     /smart-sdd adopt
```

```
<select>
```

```
     --scope core
```

/smart-sdd

```
init -> add -> constitution -> specify -> plan -> tasks
     -> analyze -> implement -> verify -> merge
```

sdd-state.md

add

→→

[preserved]

[new]

[updated]

gui

realtime

ai-assistant

spec-draft.md

---

verify-report.md

runtime: [specific method]
RUNTIME_BLOCKED: [reason]
RUNTIME_DELEGATED: [reason]

RUNTIME_BLOCKED

npm run lint

37 -> 52 (+15 auth/RBAC tests)
37 -> 37 (no new tests)

RUNTIME_BLOCKED

runtime: [method] runtime: curl POST /auth/login → 200 with token

RUNTIME_BLOCKED: [reason] RUNTIME_BLOCKED: requires 100 concurrent connections (load testing infrastructure)

RUNTIME_DELEGATED: [reason] RUNTIME_DELEGATED: drag-and-drop file upload requires manual browser interaction

RUNTIME_BLOCKED: [actual infrastructure reason] --- unit test exists as proxy

```
runtime
```

sdd-state.md

sdd-state.md

→

→

→

→→

≤

```
pipeline F00N --start implement
pipeline F00N --start plan
pipeline F00N --start specify
add --to F00N→--start specify
add "description"→
```

```
Classification: Spec Gap
Impact: FR-004 (estimation), SC-003 (deduction flow), SC-006 (concurrency)
Files: ~2 (budget.guard.ts, Lua script)

Proposed: pipeline F004 --start specify
  -> FR-004 update + SC cascade

Alternative: New Feature F008 (T2 scope)

Proceed?
```

/smart-sdd adopt

roadmap.md

/addsdd-state.md

```
→→
→
```

demos/F004-token-budget.sh

/add
demos/DG{N}-{scenario}.sh

--ci
sdd-state.md

--ci
→→

demos/DG{N}-{scenario}.sh
--ci

sdd-state.md

→→→

/domain-extend

rate-limiting

shared/domains/concerns/rate-limiting.mdreverse-spec/domains/concerns/rate-limiting.md
smart-sdd/domains/concerns/rate-limiting.md

_taxonomy.md

_resolver.md

```
/domain-extend extendspecs/domains/




--skill
```

org-convention.md

domain-custom.md

shared/domains/{axis}/{name}.md

reverse-spec/domains/{axis}/{name}.md

smart-sdd/domains/{axis}/{name}.md

app/models/domains/concerns/

_schema.md_schema.mdauth.md

/domain-extend validate

domains/_taxonomy.md

_resolver.md

specify

_resolver.md

```
→ai-assistantguirealtimeexternal-sdkauth
→public-apimicroservicehttp-apiauthresilience
→cli
→reverse-spec
→/smart-sdd init
```

+migration+compliance/domain-extend

_resolver.md

prototype + solo

mvp + small-team

production + large-team

sdd-state.md

t('key')

grep

gui + realtime

≠ ~
~
~
~
~
~
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~
~
~
≠ ~
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~
~

```
Phase 1 (Week 1): SC verification -> Foundation Gate -> Source Reference
Phase 2 (Week 1-2): SBI Cross-Check -> Playwright fallback chain
```

```
Phase 3 (Week 2): UX Contract -> Verify Progress -> SC Matrix
Phase 4 (Week 2-3): i18n Lint -> SDK Contract -> Type Trust
Phase 5 (Week 3): BLOCKING gates -> Inline Execution -> Stop Hook
Phase 6 (Post-release): Context Budget -> Lazy Loading -> Reset Protocol
Phase 7 (Ongoing): 3 Principles -> Domain Module Loading Protocol
```

MANDATORY---You MUST NOT skip it

~

navbarPosition: 'left''top'

MANDATORYCRITICALMANDATORY

tasks.md

→
→
→

→

research.mddata-model.mdcontracts/

CRITICAL--autoCALIBRATIONROUTINE

→→→→

```
/clear/smart-sdd pipeline F00N --start verify
```

```
# Input instruction (weak):
"You must verify all 4 phases before merge."

# Output verification (strong):
"Read sdd-state.md verify_progress. If any phase != complete,
 BLOCKING: 'Verify incomplete. Complete phases before merge.'"
```

```
# After each verify phase:
Write to sdd-state.md:
  phase_0_preflight: complete
  phase_1_build: complete
  phase_2_ui: in_progress (SC-003 remaining)
  phase_3_regression: pending

# On resume: read file, continue from SC-003
```

```
# Normal exit: AskUserQuestion with review
# Abnormal exit (any reason):
"[command] executed for [FID].
 Type 'continue' to review the results."

# Never: silent exit with no user guidance
```

```
# In pipeline.md specify section:
"HARD STOP: AskUserQuestion. If response empty -> re-ask."

# In pipeline.md plan section:
"HARD STOP: AskUserQuestion. If response empty -> re-ask."

# In pipeline.md tasks section:
"HARD STOP: AskUserQuestion. If response empty -> re-ask."

# Same text 30+ times. Intentional. Not a code smell.
```

/smart-sdd

```
git clone https://github.com/coolhero/spec-kit-skills.git
cd spec-kit-skills
./install.sh      # creates symlinks -> ~/.claude/skills/
```

```
/reverse-spec --help     # Should show command help
/smart-sdd status        # Should show status or ask to init
/domain-extend browse    # Should show domain module overview
```

```
# 1. Initialize project
/smart-sdd init "AI-powered knowledge base with provider abstraction"

# 2. Define first Feature
/smart-sdd add "knowledge base CRUD with file upload"
# -> 6-step Brief consultation
# -> Review and approve

# 3. Run pipeline
/smart-sdd pipeline F001
# -> constitution -> specify -> plan -> tasks
# -> implement -> verify (4 phases)
# -> At each step: review and approve

# 4. Add more Features
/smart-sdd add "multi-provider chat with streaming"
/smart-sdd pipeline F002
# F002 automatically sees F001's entities and APIs
```

verify-report.md

```
Have existing code?
  No  -> /smart-sdd init -> add -> pipeline        (New Project)
  Yes -> What is your goal?
         Understand only    -> /code-explore          (Explore)
         Document only      -> /smart-sdd adopt       (Adoption)
         Add features       -> adopt -> add -> pipeline (Extend)
         Rebuild (same stack) -> /reverse-spec -> pipeline (Rebuild)
         Rebuild (new stack)  -> /reverse-spec --stack new (Migrate)

Need custom domain modules?
  -> /domain-extend detect -> extend
```

/smart-sdd add → pipeline

```
my-project/
  specs/
    _global/
      roadmap.md             <- Feature dependency graph
      entity-registry.md     <- All data models
      api-registry.md        <- All API contracts
      sdd-state.md           <- Pipeline state + Domain Profile
      constitution-seed.md   <- Project principles
    explore/                 <- code-explore output
      orientation.md
      traces/
    001-auth/                <- Per-Feature artifacts
      pre-context.md         <- Context from analysis
      spec-draft.md          <- Initial spec (from reverse-spec)
      spec.md                <- Final spec
      plan.md                <- Architecture
      tasks.md               <- Implementation tasks
    002-task-crud/
      ...
  .specify/
    memory/
```

```
      constitution.md        <- Active constitution
```

```
/code-explore [path]
/code-explore trace "topic"
/code-explore synthesis
/reverse-spec [path]
/smart-sdd init
/smart-sdd add
/smart-sdd pipeline [FID]
/smart-sdd adopt
/smart-sdd status
/smart-sdd reset [FID]
/domain-extend browse
/domain-extend detect
/domain-extend extend
/domain-extend validate
```

/init

realtimewebsocketauthrbacdata-integrity

realtimeauthdata-integrity

app.use(cors())authMiddleware

```
+migration

urls.py→→→→→

run_in_executor


@login_requiredDepends(get_current_user)→


message-queueemailpush-notification→E003sdd-state.md
```

→→/specify/plan

→→→

+migration

↔

+migration

+migration+migration+migration

```
/smart-sdd init "AI-powered knowledge base with provider abstraction"
/smart-sdd add "knowledge base CRUD"
/smart-sdd pipeline F001
/smart-sdd add "multi-provider chat with streaming"
/smart-sdd pipeline F002
```

```
/reverse-spec /path/to/legacy-app
/smart-sdd init --from-reverse-spec
/smart-sdd pipeline F001
# ... continue in dependency order
```

```
/smart-sdd adopt /path/to/existing-app
```

```
/code-explore /path/to/reference
/code-explore trace "how does context management work"
/code-explore trace "how does provider abstraction work"
/code-explore synthesis
/smart-sdd init --from-explore
/smart-sdd add --from-explore C001
/smart-sdd pipeline F001
```

```
# Step out of the pipeline temporarily
/code-explore . --no-branch
/code-explore trace "how does the original handle concurrent file uploads"
# -> Trace reveals: source uses a queue with deduplication

# Return to the pipeline with this understanding
/smart-sdd pipeline F003 --continue
```

```
/domain-extend detect /path/to/code
# -> "Detected uncovered pattern: video-encoding"
/domain-extend extend concerns/video-encoding
# -> Creates 3-file module set
/domain-extend validate
# -> Schema compliance check passes
```

/clearsdd-state.md

sdd-state.md

add --to

add --to F003 "add PUT /users/:id"→→→→→→

→
→
→

→

```
Framework: react, fastapi
```

foundations/react.mdfoundations/fastapi.md

```
SKILL.md        <- Always loaded (~200 lines): routing + mandatory rules
commands/       <- Loaded on-demand (only the invoked command)
reference/      <- Supporting rules and schemas
domains/        <- Modular expertise (loaded per profile)
```

spec-kit-skills

~

/smart-sdd init

```
Interfaces:  [http-api, gui]
Concerns:    [auth, multi-tenancy, resilience, observability,
              realtime, stream-processing, external-sdk,
              token-budget*, prompt-guard*]
Archetype:   [ai-gateway*]
Foundation:  typescript-nestjs
Context:     greenfield | mvp x small-team

# * = project-local custom modules (not in built-in library)
```

```
        >>
        <
```

```
/domain-extend extend
```

```
/initspecify/addplanimplement
```

```
     →→→→
     →→→
```

```
     extractUserKey()
     request.cacheHit                      cacheHit: false
```

```
--start specify→→→→
```

```
failedreleased
```

$=$

$\rightarrow$

$\rightarrow$

$\sim$

$\neq$

$\rightarrow$

$\rightarrow\rightarrow\rightarrow\rightarrow$

FILE-MAP.md

```
    README.mdREADME.ko.md
    ARCHITECTURE-EXTENSIBILITY.md.ko.md
    SCENARIO-CATALOG.md.ko.md
    FILE-MAP.md
    CLAUDE.md
    history.md
    lessons-learned.md
    PLAYWRIGHT-GUIDE.md
```

```
code-explore [path]→trace→synthesis
code-explore --scope src/auth
code-explore Acode-explore B
code-explore .
code-explore .

init "..."→add→pipeline
code-explore→init --from-explore
init spec.mdadd requirements.md
pipeline F003pipeline --tier 1
pipeline --continue→init

adopt
adopt→add→pipeline
adopt→pipeline
adopt --scope services/api
adopt→pipeline --migration

reverse-spec→init --from-reverse-spec→pipeline
code-explore→reverse-spec --from-explore
reverse-spec→pipeline --tier 1


pipeline F001 --start verify
pipeline F001 --step specify
```

```
add --to F001 "add OAuth"
addpipeline merge
pipeline F001 --step specify,plan

pipeline F001pipeline F002
pipeline F001 --step specify

status
paritycoverage

init --profile sdk-library→→
adopt→pipeline --migration
init --lang koadopt --lang ja
adopt --scope services/api
domain-extend browsebrowse concerns
domain-extend detect→extend concern "video-encoding"
domain-extend import ./docs/adr/
domain-extend customize org
domain-extend detect --from-explore
domain-extend validate
```

```
/smart-sdd init "project description"
/smart-sdd add "Feature description"
/smart-sdd pipeline F001
```

```
/reverse-spec /path/to/source
/smart-sdd init --from-reverse-spec
/smart-sdd pipeline F001
```

```
/code-explore /path/to/project
/code-explore trace "topic"
/code-explore synthesis
/smart-sdd init --from-explore
```

```
/domain-extend browse
/domain-extend detect /path/to/code
/domain-extend extend concerns/new-pattern
```

```
--lang ko|en|ja
--scope core|full
--from-explore path
--from-reverse-spec
--start step
--to FID
--tier N
--all
--adopt
--no-branch
```

_resolver.md

domain-profile-instance.md

specs/domains//domain-extend extend/domain-extend import

_resolver.md

specify

--tier

add --to

```
init
import
init --prd
specify
trace
production
```

```
Read targets:
  1. specs/_global/entity-registry.md
  2. specs/_global/api-registry.md
  3. specs/{FID}/pre-context.md
  4. specs/{FID}/spec-draft.md (rebuild only)
  5. .specify/memory/constitution.md

Domain sections loaded: S0, S1, S5, S9, A2, A3, A5
Sections discarded: S2, S3, S4, S6, S7, S8, F0-F9

Injected content:
  - Brief-to-SC mapping instructions
  - SC numbering convention
  - Measurability criteria from S1
  - Cross-concern emergent rules (from Step 3.5)
  - Scale-adjusted SC depth

Checkpoint display:
  - Feature name and ID
  - Loaded domain modules with section counts
  - Entity/API registry summary (N entities, M endpoints)
  - Pre-context summary (key requirements)

Review display:
  - Generated FRs with SC counts
  - SC measurability assessment
  - Brief-to-SC coverage (every Brief item -> at least 1 SC)
  - Entity references (new vs existing)
  - AskUserQuestion: Approve / Request modifications
```

```
Read targets:
  1. specs/{FID}/spec.md
  2. specs/_global/entity-registry.md
  3. specs/_global/api-registry.md
  4. specs/{FID}/pre-context.md (for source component mapping)
```

```
Domain sections loaded: S7 (B-1 only), F2, F3
Sections discarded: S0, S1, S2, S3, S4, S5, S6, S8, S9

Injected content:
  - Bug prevention from S7 (B-1 plan-relevant subset)
  - Foundation checklist items from F2
  - T0 Feature grouping rules from F3
  - Source Component Mapping instructions (rebuild only)
  - Integration Contract template

Review display:
  - Architecture summary (components, data flow)
  - API contracts
  - Integration Contracts (if cross-Feature dependencies)
  - Source Component Mapping (rebuild only, BLOCKING)
  - AskUserQuestion: Approve / Request modifications
```

```
Read targets:
  1. specs/{FID}/tasks.md
  2. specs/{FID}/plan.md
  3. specs/{FID}/pre-context.md (BLOCKING for rebuild)
  4. Preceding Features' interaction-surfaces.md

Domain sections loaded: S7 (B-3 subset), S6, F8
Sections discarded: S0, S1, S2, S3, S4, S5, S8, S9

Injected content:
  - Per-task bug prevention anti-patterns from S7
  - UI testing integration from S6 (when gui interface active)
  - Toolchain commands from F8
  - Source Reference Injection (BLOCKING for rebuild+GUI)
  - Visual Reference Checkpoint (BLOCKING for first UI task)
  - Interaction Surface Preservation (read inventory before
    modifying shared components)

Post-step update:
  - New entities -> entity-registry.md
  - New API endpoints -> api-registry.md
  - Interaction surfaces -> specs/{FID}/interaction-surfaces.md
```

```
Read targets:
  1. specs/{FID}/spec.md (for SC list)
  2. specs/{FID}/plan.md (for Integration Contracts)
  3. verify-phases.md (Phase 0-4 protocol)
  4. sdd-state.md verify_progress (for resume)

Domain sections loaded: S1, S3, S7 (B-4), S8, F8
Sections discarded: S0, S2, S4, S5, S6, S9, F9
```

```
Injected content:
  - SC verification checklist from S1
  - Additional verify steps from S3
  - Runtime verification strategy from S8
  - Bug regression patterns from S7
  - Toolchain commands from F8

Phase structure:
  Phase 0: Playwright preflight
  Phase 1: Build + typecheck + lint (F8 commands)
  Phase 2: Automated tests
  Phase 3: UI/Runtime verification (S8 strategy)
    3a: Launch application
    3b: SC-by-SC verification against running UI
    3c: Regression checks (previous Features still work)
    3d: Source app comparison (rebuild only, Guard 7)
    3e: User-assisted verification (if Playwright unavailable)
    3f: SC Evidence Gate (all SCs must have evidence)
  Phase 4: Cross-Feature integration (Guard 6)
```

history.md

_resolver.md

××

https://github.com/coolhero/spec-kit-skills